

AutoCodeRover : Automatic Programming Research → Spinoff → Acquisition

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Ministry of Education Tier 3 program “Automated Program Repair” 2022 – 27 <https://nus-apr.github.io/>

AutoCodeRover

Motivation

- Software engineering is becoming more automated.

Example:



Test generation

SemFix

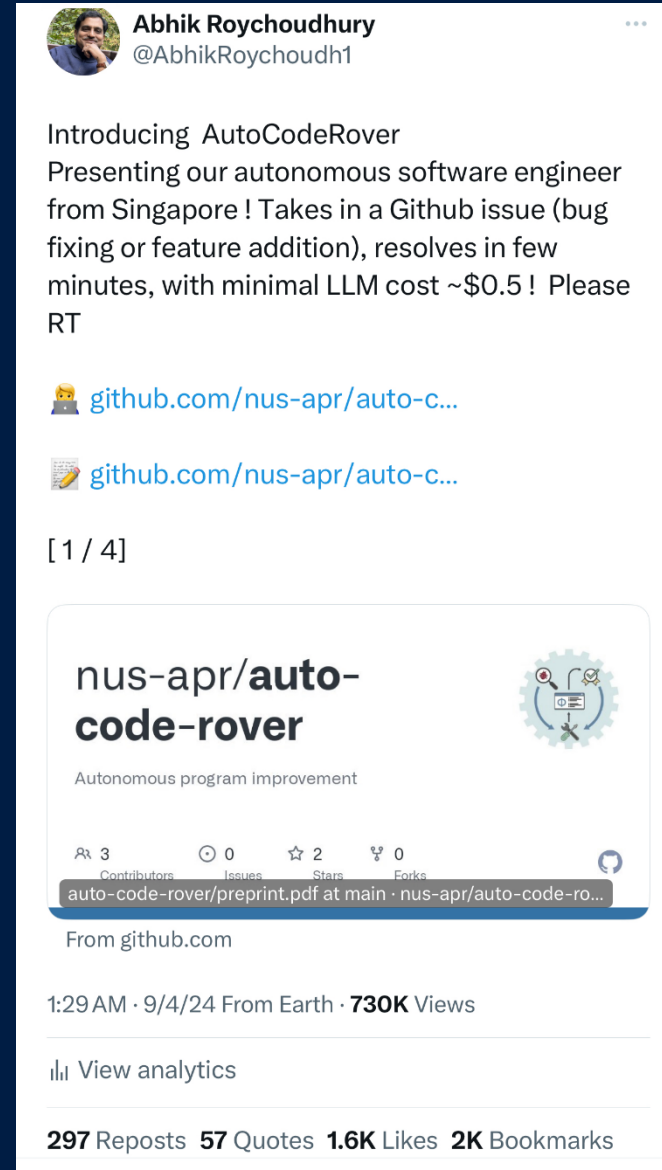
Program repair



Code generation

- Can Software maintenance be automated and autonomous?

Yes!



First step towards Autonomous Software Engineering

Automatically resolve GitHub issue **without human assistance**



Explore codebase from the **program structure**
Equip AI agent with **software debugging** capability

Initial Observations (after ICSE 2023)

SE oriented outlook on top of AI

Do not view a software project as collection of files.

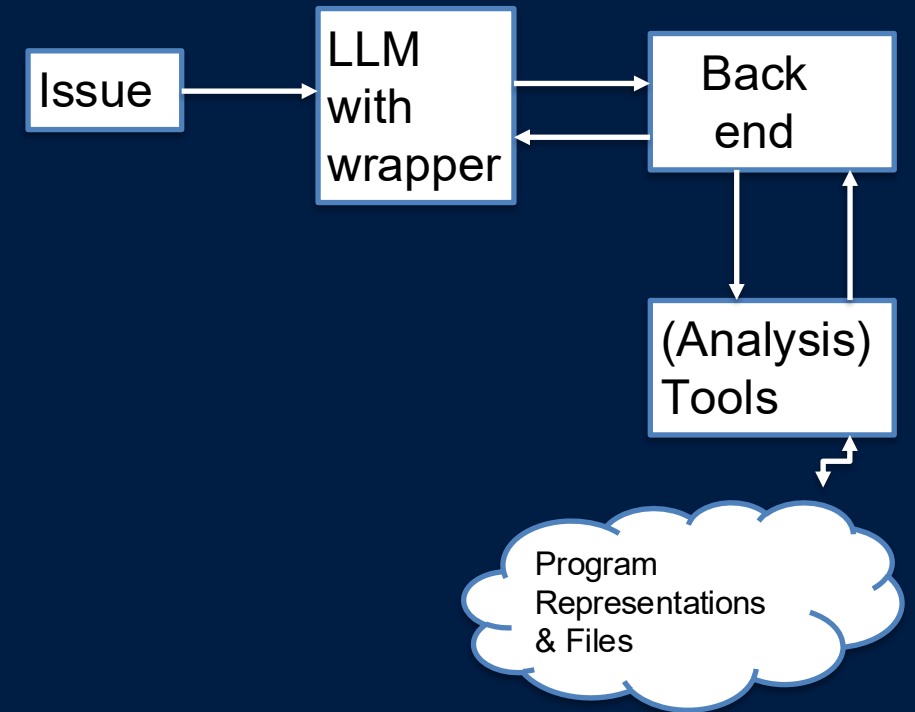
Do not see code analysis as searching for strings in files and scrolling up and down in a file.

Work on program representations : AST

Code search to exploit program structure: setting an optimal context.

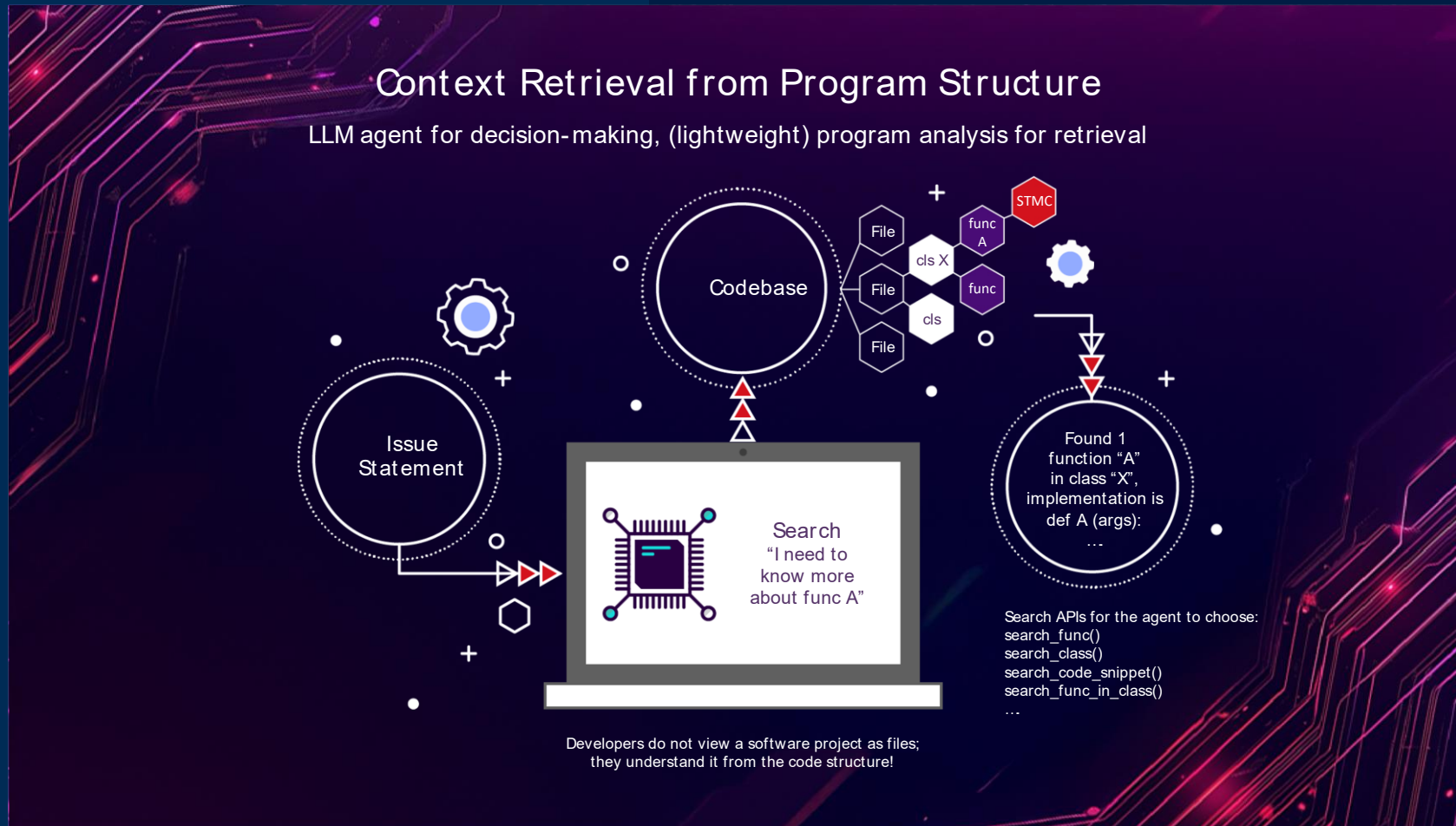
Tests, if available, can be used for fault localization and patch validation.

Program dependencies can be used to refine the code search with semantic artifacts.

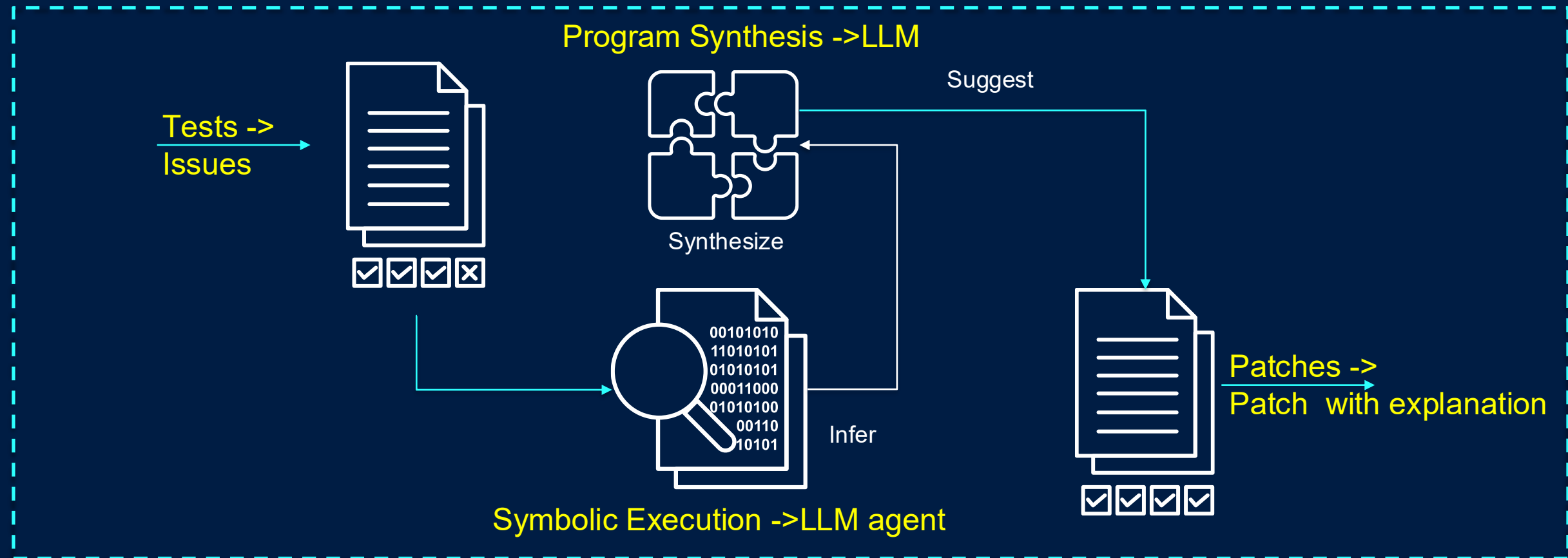


Program Structure captures intent. Extract coarse specs from structure for autonomous SE

Core Approach : Program Representations



SPEC. INFERENCE. –Research / Innovation



Program Structure captures intent. Extract coarse specs from structure for autonomous SE

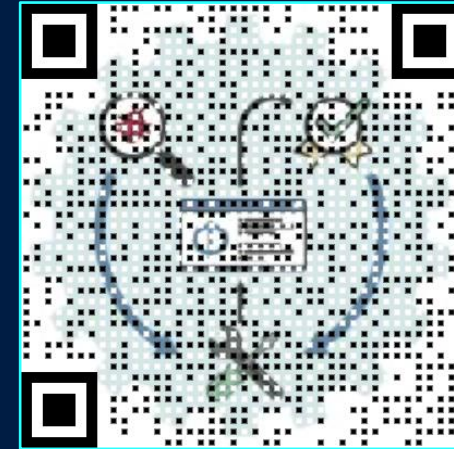
AutoCodeRover

AutoCodeRover: Autonomous Program Improvement

Yuntong Zhang, Haifeng Ruan, Zhiyu Fan, Abhik Roychoudhury

ACM International Symposium on Software Testing and Analysis (ISSTA) 2024.

arXiv:2404.05427, Publicly released 8th April 2024



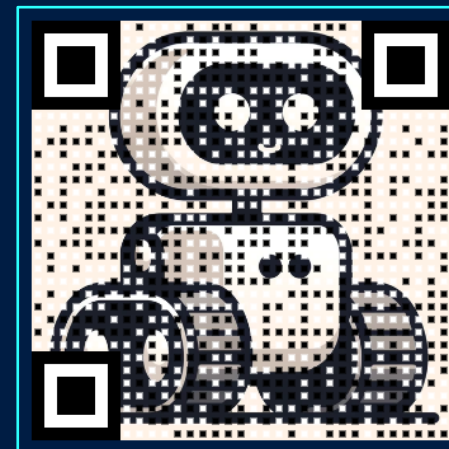
auto-code-rover

Public

A project structure aware autonomous software engineer aiming for autonomous program improvement. Resolved 15.95% tasks in full SWE-bench

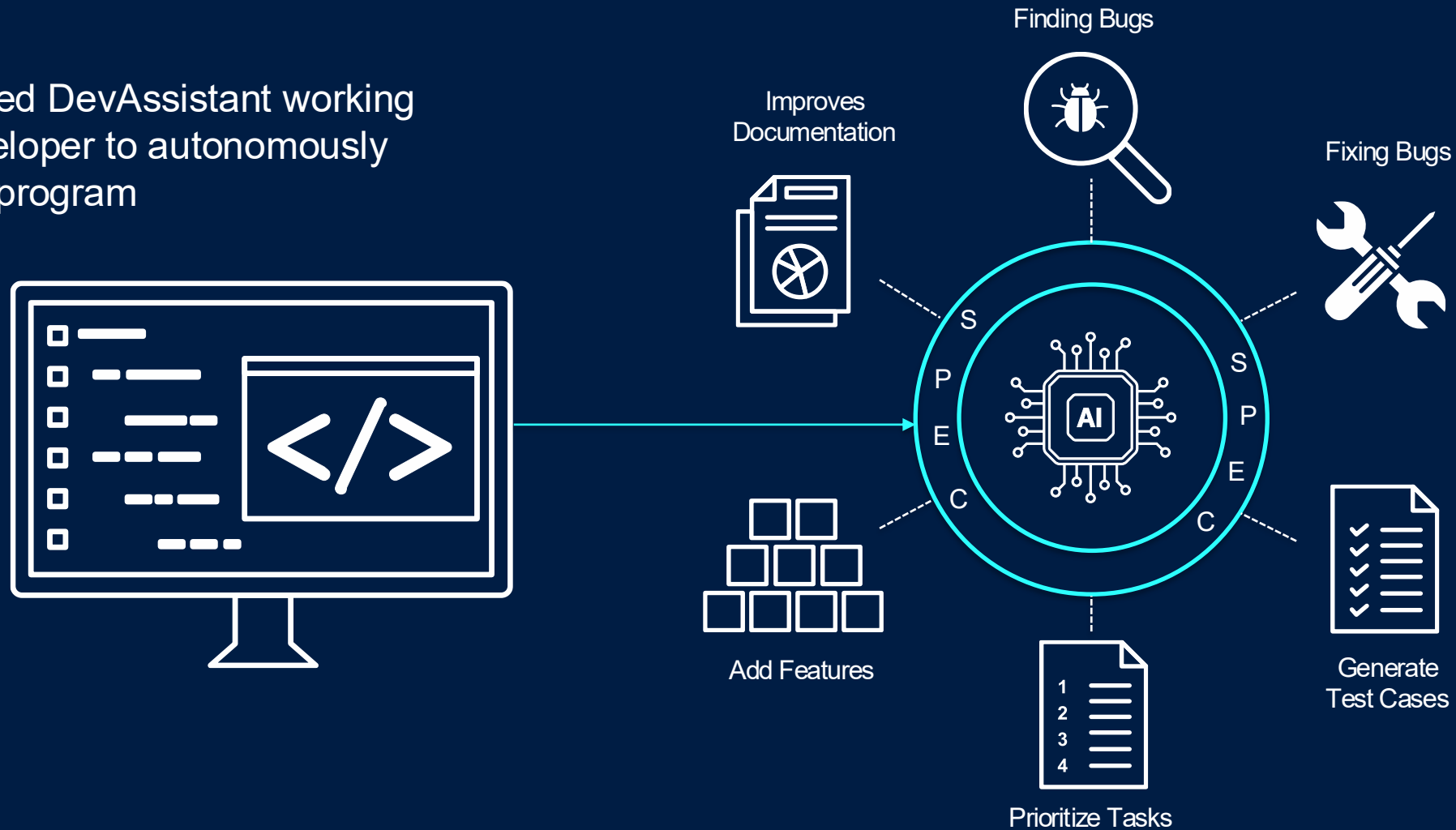
● Python ☆ 2.3k 🍴 225

<https://autocoderover.dev>



AI Software Engineer : Draft 1, Vision

An AI powered DevAssistant working with the developer to autonomously improve the program



Acquisition



WHAT'S HAPPENING

AutoCodeRover, an advanced AI agent platform transforming software development, has been acquired by Sonar, a leader in code quality and security.

WHAT'S THE TECHNOLOGY

Developed by Professor **Abhik Roychoudhury** and his team at the National University of Singapore, AutoCodeRover leverages state-of-the-art Large Language Models (LLMs) to **automate essential tasks** such as debugging, code refactoring, and issue remediation.

WHAT'S GREAT ABOUT IT

AutoCodeRover ensures optimal flexibility for users, accelerates development cycles, improves code quality, reduces cost and drives innovation.

By automating routine tasks, AutoCodeRover enables developers to dedicate more time to innovation and creative problem-solving, accelerating the delivery of high-quality applications!

Brief Lessons

Spinoff formation Traditional Wisdom – may not always apply to your research, and the ecosystem you are in

- Team Bonding
- Know your customer
- Minimum Viable Product
- Identifying pain points leading to Business Model

Brief Lessons

May need to modified or changed depending on your ecosystem

- Team Bonding

[Interesting story, let me speak about it later]

- Know your customer – may be too hard in your R&D ecosystem

[Know someone instead who knows your customer – ecosystem specific]

- Minimum Viable Product

[May need a demonstration of a more powerful capability, but not a polished product]

- Identify pain points leading to a Business Model

[May be a very fast-moving space such as AI/tech , keeping your offering moving is supreme.]

Research - Innovation – Acquisition

Well-known factors about deep tech startups

Innovation cycles are getting shorter.

The technology research is not enough ...

Know your customer ...

Team bonding ...

But what did I intervene on (preserving energy)

Inspiration



Research Quality



Ground Reality



These are perhaps the only aspects on which I focused and intervened

Closed vs. Open Innovation

Ideas

Execution

Innovator

Partner Network

Investors

Marketing

Customers

Disruptive Innovation

Ideas

Execution

Innovator

Partner Network

Investors

Marketing

Customers

Disruptive Innovation

2015-6 NUS
Teaching
Infosec

Security
testing tools
from Google

Work with user
community

2019-20
Fuzzbench by
Google OSS
for Innovation

Community
building

Open-source
innovation in
security testing

**The Innovator's Dilemma,
Clayton Christensen, 1997.**



Look out

??



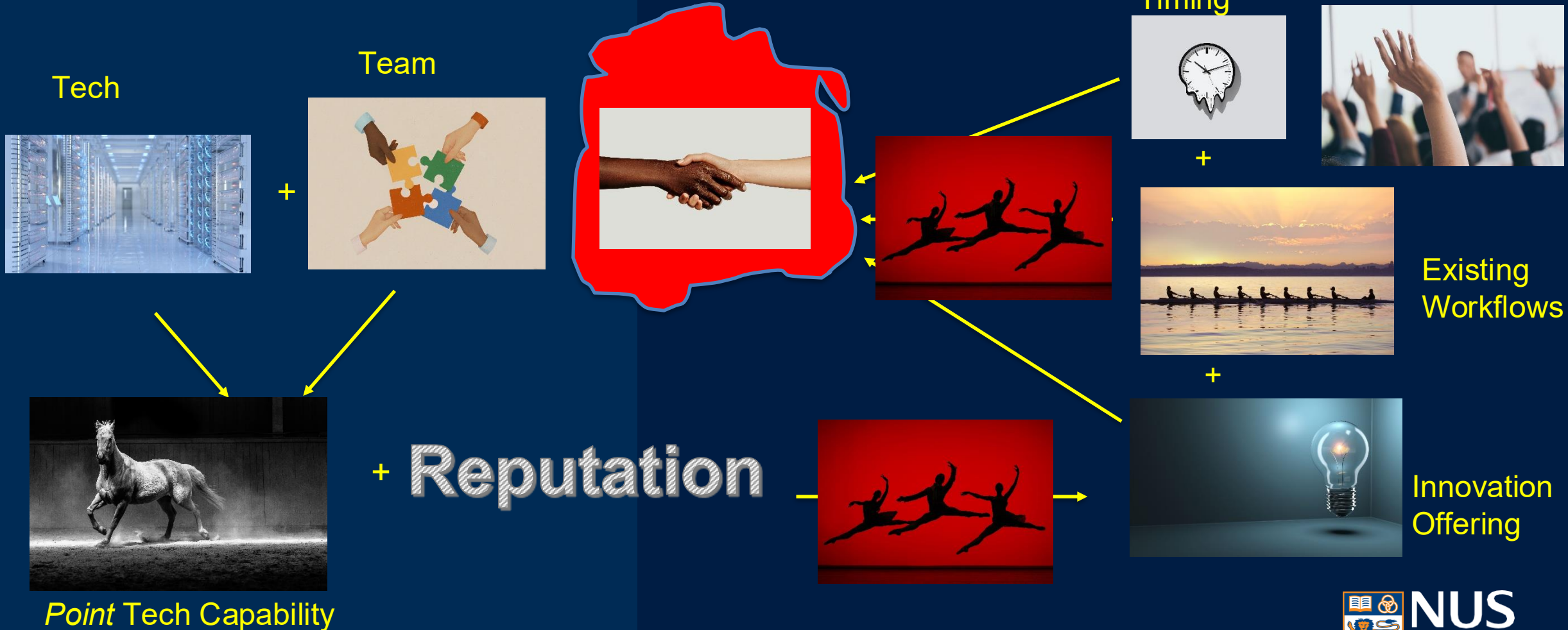
Reflect

***Past Experience in
B-U innovation***

***Innovator's Confidence
Lonely decision making***

Acquisition from the other side? RIE

Motivating factors related to deep-tech startups?



Tech



Team

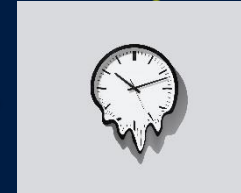


+



E

Timing



+

Stakeholder
Confidence



+

Existing
Workflows



Innovation
Offering

+ Reputation



Point Tech. Capability

Real Incident

May 18 2023: Most Influential Paper Award Talk for 2013 paper Intl. Conf on SW Engg (ICSE)

Crucial time in the
innovation cycle

Oct 24, 2023: Started solution on Large Language Model agents for SW Engg.

"Imagine all of the program analysis can be invoked autonomously"

Apr 8, 2024: Public announcement in X, Excitement around AutoCodeRover.

Feb 19, 2025: Acquisition by SonarSource announced, 9 am EST, 10 pm SGT.

Feb 20, 2025: Contacted for a group photo, realized there are no photos at all !!

Feb 21, 2025: Met for the first time outside work as a group, strong in team spirit !!



Haifeng

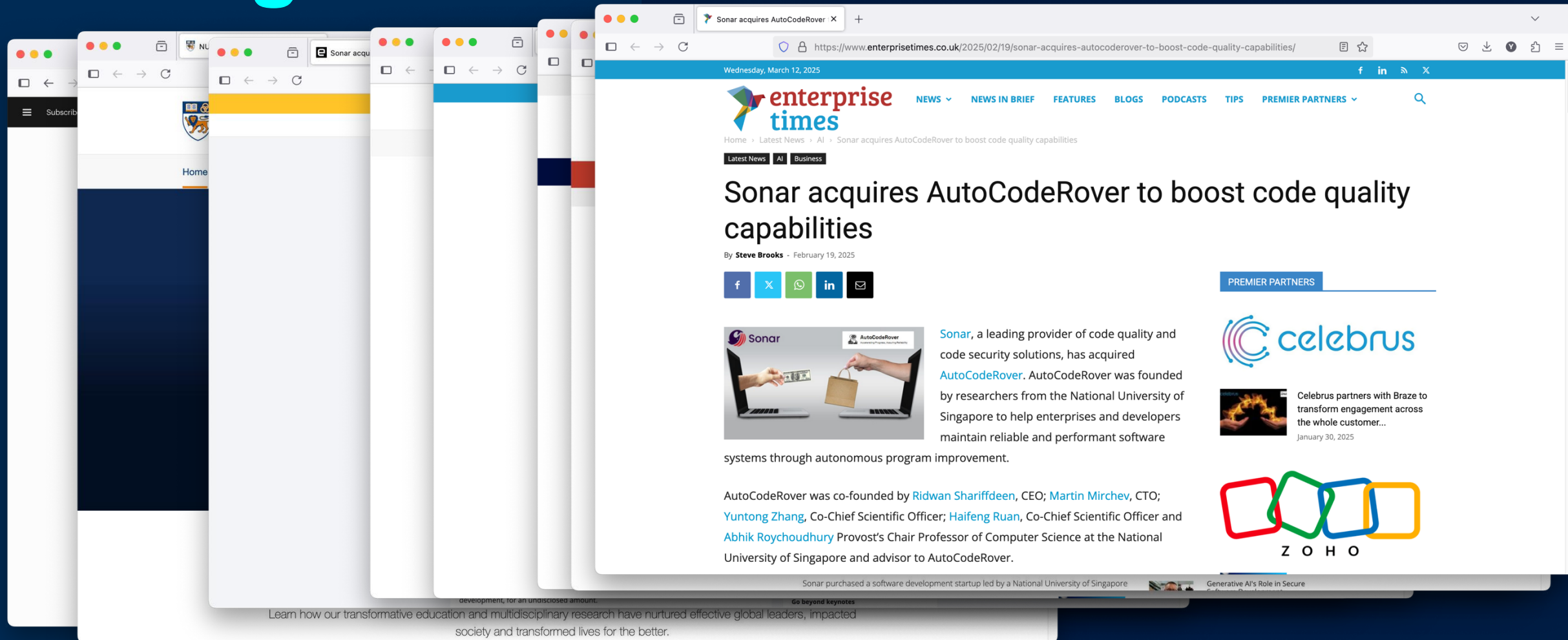
Yuntong

Ridwan

Martin

Abhik

Coverage



Experience – Deep tech innovation

RESEARCH

Disruptive tech may be an opportunity.

Connect new, old instead of obsolescence

Set own VISION !

INNOVATION

Every innovation ecosystem has challenges.

Do not go by wisdom from other ecosystems

Have agency, imagine your own MANTRA !

Stay active in research, along with innovation



Research Reflections

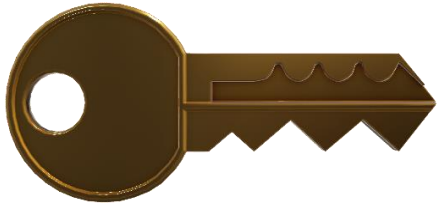
“Hello World”
1972

~50 years

Programming at Scale

Linux Kernel in 2024
~30M LoC

Automatically generated
code



~X years

Programming with Trust
Role for Verification



Cooperative Intelligence

Tech Ecosystem Development

Study the aspects of the ecosystem you are in
(for us Singapore)

- Stage of the ecosystem
- Growth potential
- What is the next step needed there?

Lot of tech R&D activity and tech R&D jobs

Focus of AutoCodeRover effort

Sonar Singapore is hiring !!

***10 R&D jobs already created.
Several others in the process.***

